Explore the network

Start here

WSL: start an exercise. Each command resets its scenario.

```
make networking      # follow a request
make operations      # find the fault
make automation      # repair through an API
make monitoring      # measure traffic and loss
```

Help: `make task`, `make hint STEP=2`, `make solution STEP=2`.

Lagoon probe (replace `lagoontransit` with `pacifictransit` for Pacific):

```
make enter NODE=probe01.bob1.lagoontransit.test
```

Inside a probe

```
ip -br addr                              # my addresses
ip route get 198.51.100.10               # IPv4 gateway and source
ip -6 route get 2001:db8:100::10         # IPv6 gateway and source
dig @198.51.100.10 data.oceanresearch.test A +time=1 +tries=1
dig @2001:db8:100::10 data.oceanresearch.test AAAA +time=1 +tries=1
ping -4 -c 3 -W 1 198.51.100.10          # IPv4 reachability
ping -6 -c 3 -W 1 2001:db8:100::10       # IPv6 reachability
curl --fail --max-time 5 -4 http://data.oceanresearch.test/
curl --fail --max-time 5 -6 http://data.oceanresearch.test/
```

Look inside a router

In WSL: enter ReefNet Edge 01.

```
make enter NODE=edge01.bob1.reefnet.test
```

For Lagoon, use `NODE=edge01.bob1.lagoontransit.test`.

SR Linux CLI

```
show interface brief
show network-instance default protocols bgp neighbor
show network-instance default protocols ospf neighbor
```

```
show network-instance default protocols bgp routes ipv4 prefix 198.51.100.0/24
show network-instance default route-table ipv4-unicast prefix 198.51.100.0/24
show network-instance default protocols bgp routes ipv6 prefix 2001:db8:100::/48
show network-instance default route-table ipv6-unicast prefix 2001:db8:100::/48
```

On ReefNet Edge 01, what do we advertise to Lagoon? Which policies apply?

```
show network-instance default protocols bgp neighbor 192.0.2.2 advertised-routes ipv4
info from running network-instance default protocols bgp
info from running routing-policy
```

Inspect and change configuration

Management access, links and return route

From the Lagoon probe, compare the IPv4 and IPv6 paths. A silent hop alone does not prove failure:

```
traceroute -n -4 -w 1 -q 1 -m 8 198.51.100.10
traceroute -n -6 -w 1 -q 1 -m 8 2001:db8:100::10
```

Router management names resolve from the operator workstation. In WSL:

```
make enter NODE=operations01.bob1.reefnet.test
```

Inside operations01:

```
getent hosts edge01.bob1.reefnet.test
ssh edge01.bob1.reefnet.test
```

Compare the ReefNet-Lagoon link at both ends. On ReefNet Edge 01:

```
show interface ethernet-1/3 detail
```

On Lagoon Transit:

```
show interface ethernet-1/1 detail
show interface ethernet-1/2 detail
```

Lagoon `ethernet-1/1` faces ReefNet. `ethernet-1/2` faces the probe.

Check the return route on the customer edge:

```
make enter NODE=edge01.bob1.oceanresearch.test
```

```
show network-instance default route-table ipv4-unicast prefix 203.0.113.0/25
show network-instance default route-table ipv4-unicast prefix 203.0.113.128/25
show network-instance default route-table ipv6-unicast prefix 2001:db8:a::/64
show network-instance default route-table ipv6-unicast prefix 2001:db8:b::/64
```

Change the Lagoon export policy from the CLI

After the investigation, inspect both the group override and BGP's inherited policy:

```
info from running network-instance default protocols bgp group lagoon-v4
info from running network-instance default protocols bgp
```

Stage the scoped change. It removes the group's override so it inherits `EXPORT-BGP`.

```
enter candidate
delete / network-instance default protocols bgp group lagoon-v4 export-policy
diff
```

Commit only if `diff` shows that single removal. Then verify the running config and test from the probe.

```
commit now
info from running network-instance default protocols bgp group lagoon-v4
```

Use `discard now` to cancel uncommitted edits. To restore the committed before-state, set the original attachment:

```
enter candidate
set / network-instance default protocols bgp group lagoon-v4 export-policy [ BLOCK-CUSTOMER-V4 ]
diff
commit now
```

Read and change through APIs

Read through an API

In WSL: enter the operator workstation.

```
make enter NODE=operations01.bob1.reefnet.test
```

Run the commands below **inside operations01**.

Find a device in NetBox

```
TOKEN=$(cat /state/netbox-token)
NETBOX_API=http://netbox.bob1.reefnet.test:8080/api
curl -fsSG -H "Authorization: Bearer $TOKEN" \
  "$NETBOX_API/dcim/devices/" \
  --data-urlencode name=edge01.bob1.reefnet.test |
  jq '{count, devices: [.results[] | {name, primary_ip4}]}'
```

Expect one device. Check its name and primary IP. Keep the token private.

Read the BGP configuration with gNMI

```
BGP_PATH='/network-instance[name=default]/protocols/bgp'
gnmic -a edge01.bob1.reefnet.test:57400 \
  -u admin -p 'NokiaSrl1!' --skip-verify --encoding json_ietf \
  get --type config --path "$BGP_PATH"
```

Change a BGP policy with gNMI

Still inside operations01: read the affected group and note its current policy.

```
BGP_PATH='/network-instance[name=default]/protocols/bgp'
GROUP="$BGP_PATH/group[group-name=lagoon-v4]"
gnmic -a edge01.bob1.reefnet.test:57400 \
  -u admin -p 'NokiaSrl1!' --skip-verify --encoding json_ietf \
  get --type config --path "$GROUP"
```

Remove the group's policy override

After diagnosis: the group uses `BLOCK-CUSTOMER-V4`, causing the fault. Its parent uses `EXPORT-BGP`. Delete the override to inherit that policy.

```
gnmic -a edge01.bob1.reefnet.test:57400 \
  -u admin -p 'NokiaSrl1!' --skip-verify --encoding json_ietf \
  set --delete "$GROUP/export-policy" --dry-run
```

Check the target and path. `--dry-run` previews the request but does not validate it on the router. When it matches your diagnosis, repeat without `--dry-run` to apply it:

```
gnmic -a edge01.bob1.reefnet.test:57400 \
  -u admin -p 'NokiaSrl1!' --skip-verify --encoding json_ietf \
  set --delete "$GROUP/export-policy"
gnmic -a edge01.bob1.reefnet.test:57400 \
  -u admin -p 'NokiaSrl1!' --skip-verify --encoding json_ietf \
  get --type config --path "$GROUP"
```

Verify routes and service from both probes. Use `exit` to return to WSL. Change and rollback: `make solution SCENARIO=automation STEP=3`.

Measure and experiment

Watch traffic and loss

In WSL: choose one load. Wait at least 30 seconds at each level.

```
make traffic-40mbit
make traffic-60mbit
make traffic-75mbit
make traffic-10mbit
make traffic-diagnostics    # inspect counters and receiver loss
```

Grafana: http://localhost:3000, folder **ReefNet / BOB1**. Check the interface, direction and time window.

In WSL: `make enter NODE=edge01.bob1.lagoontransit.test` In its router CLI:

```
info from state interface ethernet-1/1 statistics
```

Compare `out-discarded-packets` at two times.

Leave with `quit`. In WSL: `make enter NODE=service01.bob1.oceanresearch.test` Inside the service node:

```
tail -n 15 /tmp/iperf-server.log   # received Mbit/s and loss
```

Control the experiment

In WSL: run `make traffic-10mbit`, then `make test`. Start only when all four results pass. Try one fault at a time.

Experiment	Inject	Restore
Link failure	`make fault-link`	`make clear-link`
Routing failure	`make fault-routing`	`make clear-routing`

Observe, restore, then repeat `make test` after convergence. Investigation: `make hint SCENARIO=monitoring STEP=4` (link), `STEP=5` (routing).

Short burst

```
make traffic-stop           # stop continuous traffic first
```

For the rate-window comparison, wait five minutes. Then:

```
make traffic-burst-40mbit   # stops after five seconds
make traffic-status         # current load
```

Queries: `make hint SCENARIO=monitoring STEP=6`.

Stop and repeat

Keep notes on your findings. `make down` removes the lab and discards router changes.

Scenario commands and `make reset` recreate the initial state. `make clean` also deletes lab state and NetBox data.